PROJECT S.E.E.S. — SCALE RC VEHICLE PROTOTYPE

OPTION 3 BLUEPRINT V1.0

1:10 SCALE WORKING DEMONSTRATOR GUIDE FOR NARSIMHA & TEAM

Innovators: Durgam.Noushik (Lead), Narsimha (Hardware), B.Tejesh

Target Showcase: Telangana Future Innovators Challenge (TFIC) & DSO

Chassis: 1:10 Scale Toy/RC Car Chassis (₹500–₹1,000)

Safety Standard: 100% Non-Hazardous Low-Voltage (< 6V DC) Operation

1. Photographic Component Blueprint (How it Looks Assembled)

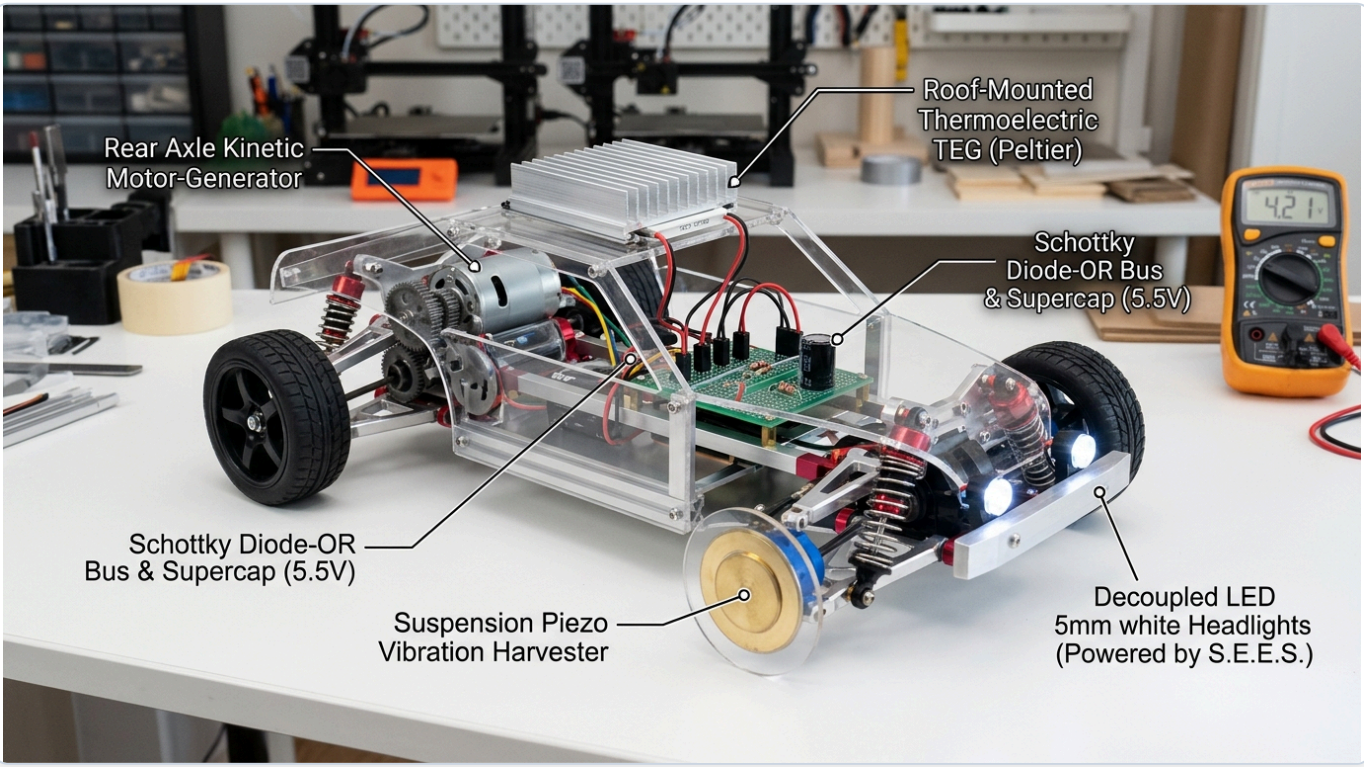


Figure 1.0: Real Component Placement inside 1:10 Scale RC Car Chassis

2. Component Layout & Vehicle Mapping

Vehicle Location	Installed Harvester	Physical Mechanism	Electrical Output
Rear Axle	12V DC Motor-Generator	Geared to rear axle; spins as toy car rolls across floor/table	**2.5V – 4.5V DC (Surge)**
Roof / Hood	TEC1-12706 Seebeck TEG + Heatsink	Warm cup or heated metal plate placed on top simulates exhaust heat	**1.5V – 1.8V DC (Baseload)**
Front Suspension	Piezoelectric Ceramic Disc	Bonded to front axle spring; vibrates when rolling over obstacles	**AC Spikes $\to$ DC Bridge**
Interior Cockpit	Schottky Diode-OR Bus + 5.5V Supercap	3× 1N5819 diodes block back-feeding; buffer stores all charge	**2.40V Regulated Bus**
Front Bumper	2× 5mm White/Green LED Headlights	Connected through cockpit toggle switch as the auxiliary load	**Runs on 0W Battery Power!**

3. Step-by-Step Build Instructions for Narsimha

STEP 1: Rear Axle Kinetic Coupling

Take the toy car's rear axle. Glue a small gear or rubber pulley to the axle. Mount the 12V DC motor inside the rear chassis so its shaft meshes with that gear. Whenever the car rolls forward on the floor, the spinning wheels turn the motor shaft, immediately generating **2.5V to 4.5V DC**!

STEP 2: Roof/Hood Thermal Module Mounting

Mount the TEC1-12706 Peltier module flat on the roof or hood with its aluminum heatsink underneath. During a demonstration, placing a warm cup of water, a heated metal washer, or warm water on top instantly generates **1.5V DC continuous baseload current**.

STEP 3: Suspension Piezo Installation

Affix the round brass piezoelectric disc onto the front suspension arm or spring. When the toy drives over books, pens, or floor bumps, the vertical bouncing flexes the disc, generating electrical charges through the small bridge rectifier.

STEP 4: Cockpit Schottky Diode-OR Bus & Headlights

Mount a small mini-breadboard inside the car interior. Wire each generator's positive lead through its own **1N5819 Schottky diode** (stripes pointing toward the supercapacitor). Connect the supercapacitor buffer directly to the front LED headlights through a small dashboard slide switch.

4. The 30-Second Live Demonstration (How to Win Competitions)

THE LIVE DEMO TRICK THAT WILL BLOW TFIC JUDGES & THE DSO AWAY:

1. Place the scale car on the judges' table with headlights OFF.
2. Open the toy car's battery compartment and show that there are **NO internal batteries** installed!
3. Push the car forward across the table (Kinetic) and touch a warm cup to the roof (Thermal).
4. Flip the headlight switch $\longrightarrow$ **THE BRIGHT LED HEADLIGHTS TURN ON AND STAY ILLUMINATED!**
5. Tell the judges: "*Sir, this vehicle's headlights are powered 100% by ambient motion and heat. In a real electric vehicle, this saves 15% to 25% of the main traction battery!*"

Why This Beats a Real Car:

- **100% Legal & Safe:** Zero high-voltage risk, zero fuel or exhaust hazards, complies with all school and government competition safety rules.
- **Completely Portable:** Fits easily in your school bag; you can carry it into the DSO office, classrooms, and exhibition halls.
- **Budget Friendly:** An inexpensive toy car chassis costs ₹500–₹1,000, and all the electronics come right from your ATL lab!